

SIMATS Online Programming Lab

PYTHON PROGRAMMING

KUMARAVEL D
192319019RD

Class Work(30.07.26)

Back

QUESTIONS

Q1

Sum & Average - Tuple

10 marks

Q2

Lucky Number

10 marks

Q3

Frequency of Each Element - List

10 marks

Q4

Element Exists - Tuple

10 marks

Q5

Maximum Element - Tuple

10 marks

Q6

Missing Number - Tuple

10 marks

Q7

Remove Duplicate Elements - Tuple

10 marks

Q8

Majority Element - List

10 marks

Q9

Peak Element - List

10 marks

Q10

Second Largest and Second Smallest - Tuple

10 marks

10/10 answered

Q1 Sum & Average - Tuple

10 marks

Write a Python program to find the sum and average of the elements in a tuple.

INPUT:
10 20 30 40 50
OUTPUT:
150
30.0

Your Code

```
1 t = tuple(map(int, input().split()))
2 s = sum(t)
3 avg = s / len(t)
4
5 print(s)
6 print(f"{avg:.1f}")
```

Input

10 20 30 40 50

Output

150
30.0

Visible Test Cases

Test Case 1

10.00 / 10 marks

Submitted

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10 marks



Q7

Remove Duplicate Elements - Tuple

10 marks



Q8

Majority Element -List

10 marks



Q9

Peak Element - List

10 marks



Q10

Second Largest and Second Smallest - Tuple

10 marks



10/10 answered

Q2 Lucky Number

10 marks

Write a recursive Python function to determine whether a given number is a Lucky Number.

A Lucky Number is a number whose sum of digits is reduced repeatedly until a single digit is obtained, and the final digit is 7.

INPUT1:
34
OUTPUT1:
Lucky Number

INPUT2:
123
OUTPUT2:
Not a Lucky Number

Your Code

```
1 def is_lucky_number(n):
2     if n < 0:
3         n = -n
4     if n < 10:
5         return "Lucky Number" if n == 7 else "Not a Lucky Number"
6     else:
7         s = 0
8         while n > 0:
9             s += n % 10
10            n //= 10
11        return is_lucky_number(s)
12 num = int(input().strip())
13 print(is_lucky_number(num))
```

Input

34

123

Output

Lucky Number

Visible Test Cases

Test Case 1



Test Case 2



10.00 / 10 marks

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Q7

Remove Duplicate Elements - Tuple

10 marks

Q8

Majority Element -List

10 marks

Q9

Peak Element - List

10 marks

Q10

Second Largest and Second Smallest - Tuple

10 marks

10/10 answered

Q3

Frequency of Each Element - List

10 marks

Write a Python program to find the frequency of each element in an array using List.

INPUT:
1 2 3 2 4 1 5 2
OUTPUT:
1 = 2
2 = 3
3 = 1
4 = 1
5 = 1

Your Code

```
1 arr = list(map(int, input().split()))
2
3 freq = {}
4 for x in arr:
5     freq[x] = freq.get(x, 0) + 1
6
7 for k in sorted(freq.keys()):
8     print(f"{k} = {freq[k]}")
```

Input

1 2 3 2 4 1 5 2

Output

```
1 = 2
2 = 3
3 = 1
4 = 1
5 = 1
```

Visible Test Cases

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Missing Number - Tuple

10 marks

Q7

Remove Duplicate Elements - Tuple

10 marks

Q8

Majority Element -List

10 marks

Q9

Peak Element - List

10 marks

Q10

Second Largest and Second Smallest - Tuple

10 marks

10/10 answered

Q4 Element Exists -Tuple

10 marks

Write a Python Program to Check Whether an Element Exists in a Tuple

INPUT:
10 20 30 40 50
30
Output 1
Element found
INPUT2:
10 20 30 40 50
25
Output 1
Element not found

Your Code

```
1 a = list(map(int, input().split()))
2 x = int(input())
3 t = tuple(a)
4 print("Element found" if x in t else "Element not found")
```

Input

10 20 30 40
30
10 20 30 40 50
25

Output

Element found

Visible Test Cases

Test Case 1

Test Case 2

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Remove Duplicate Elements - Tuple

10 marks

Q8

Majority Element -List

10 marks

Q9

Peak Element - List

10 marks

Q10

Second Largest and Second Smallest - Tuple

10 marks

10/10 answered

Q5

Maximum Element - Tuple

10 marks

Write a Python program to find the largest element in a tuple.

INPUT:
12 45 8 67 23
OUTPUT:
67

Your Code

```
1 t = tuple(map(int, input().split()))  
2 print(max(t))
```

Input

12 45 8 67 23

Output

67

Visible Test Cases

Test Case 1

10.00 / 10 marks

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10 marks



Q6

Missing Number - Tuple

10 marks



Q7

Remove Duplicate Elements - Tuple

10 marks



Q8

Majority Element -List

10 marks



Q9

Peak Element - List

10 marks



Q10

Second Largest and Second Smallest - Tuple

10 marks



10/10 answered

Q6

Missing Number - Tuple

10 marks

Write a Python program to find the missing number in a list containing numbers from 1 to n using a tuple.

INPUT:
1 2 3 5 6
OUTPUT:
4

Your Code

```
1 def missing_number(arr):  
2     n = len(arr) + 1  
3     total = n * (n + 1) // 2  
4     s = sum(arr)  
5     return total - s  
6 data = tuple(map(int, input().split()))  
7 print(missing_number(data))
```

Input

1 2 3 5 6

Output

4

Visible Test Cases

Test Case 1



10.00 / 10 marks

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10 marks



Q5

Maximum Element - Tuple

10 marks



Q6

Missing Number - Tuple

10 marks



Q7

Remove Duplicate Elements - Tuple

10 marks



Q8

Majority Element - List

10 marks



Q9

Peak Element - List

10 marks



Q10

Second Largest and Second Smallest - Tuple

10 marks



10/10 answered

Q7 Remove Duplicate Elements - Tuple

10 marks

Write a Python program to remove duplicate elements from a tuple.

INPUT:
10 20 30 20 40 10 50
OUTPUT:
(40, 10, 50, 20, 30)

Your Code

```
1 t = tuple(map(int, input().split()))
2 seen = set()
3 res = []
4 for x in t:
5     if x not in seen:
6         seen.add(x)
7         res.append(x)
8 print(tuple(res))
```

Input

10 20 30 40 10 50

Output

(10, 20, 30, 40, 50)

Visible Test Cases

Test Case 1



0.00 / 10 marks

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10 marks

Q6

Missing Number - Tuple

10 marks

Q7

Remove Duplicate Elements - Tuple

10 marks

Q8

Majority Element -List

10 marks

Q9

Peak Element - List

10 marks

Q10

Second Largest and Second Smallest - Tuple

10 marks

10/10 answered

Q8 Majority Element -List

10 marks

Write a Python function to check whether a list contains a majority element.

INPUT:
2 2 1 3 2 2
OUTPUT:
2

Your Code

```
1 def majority_element(arr):
2     n = len(arr)
3     cand = None
4     count = 0
5     for x in arr:
6         if count == 0:
7             cand = x
8             count = 1
9         elif x == cand:
10            count += 1
11        else:
12            count -= 1
13    if arr.count(cand) > n // 2:
14        return cand
15    return None
16 arr = list(map(int, input().split()))
17 res = majority_element(arr)
18 print(res if res is not None else "None")
```

Input

2 2 1 3 2 2

Output

2

Visible Test Cases

Test Case 1

10.00 / 10 marks

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Q7

Remove Duplicate Elements - Tuple

10 marks

Q8

Majority Element -List

10 marks

Q9

Peak Element - List

10 marks

Q10

Second Largest and Second Smallest - Tuple

10 marks

10/10 answered

Q10

Second Largest and Second Smallest - Tuple

10 marks

Write a Python program to find the second largest and second smallest elements in a list using a tuple.

INPUT:
10 20 5 40 15 30

OUTPUT:
10
30

Your Code

```
1 nums = list(map(int, input().split()))
2 largest = second_largest = float('-inf')
3 smallest = second_smallest = float('inf')
4 for x in nums:
5     if x > largest:
6         second_largest = largest
7         largest = x
8     elif largest != x and x > second_largest:
9         second_largest = x
10    if x < smallest:
11        second_smallest = smallest
12        smallest = x
13    elif smallest != x and x < second_smallest:
14        second_smallest = x
15 result = (second_largest, second_smallest)
16 print(result[1])
17 print(result[0])
```

Input

10 20 5 40 15 30

Output

10
30

Visible Test Cases

Test Case 1

0.00 / 10 marks

Run

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ENGINEERING

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PYTHON PROGRAMMING

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Sum & Average - Tuple

10 marks

✓

Q2

Lucky Number

10 marks

✓

Q3

Frequency of Each Element - List

10 marks

✓

Q4

Element Exists - Tuple

10 marks

✓

Q5

Maximum Element - Tuple

10 marks

✓

Q6

Missing Number - Tuple

10 marks

✓

Q7

Remove Duplicate Elements - Tuple

10 marks

✓

Q8

Majority Element - List

10 marks

✓

Q9

Peak Element - List

10 marks

✓

Q10

Second Largest and Second Smallest - Tuple

10 marks

✓

10/10 answered

Q9 Peak Element - List10 marks

Write a Python program to find a peak element in a list of integers using list. [A peak element is an element that is greater than or equal to its adjacent elements.]

INPUT:
13 20 4 1 0

OUTPUT:
20

Your Code

```
1 def peak_element(arr):
2     n = len(arr)
3     for i in range(n):
4         left_ok = (i == 0) or (arr[i] >= arr[i - 1])
5         right_ok = (i == n - 1) or (arr[i] >= arr[i + 1])
6         if left_ok and right_ok:
7             return arr[i]
8     return None
9 arr = list(map(int, input().split()))
10 print(peak_element(arr))
```

Input

1 3 20 4 1 0

Output

20

Visible Test Cases

Test Case 1

✓

10.00 / 10 marks

✓ Submitted

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